

An Investigation of Board Level Effects on Compact Thermal Models of Electronic Chip Packages

Jason DeVoe, Alfonso Ortega

Department of Aerospace and Mechanical Engineering and

The Center for Electronics Packaging Research

The University of Arizona

Tucson, AZ 85721

Phone: (520)-621-6787

Email: ortega@u.arizona.edu

Abstract

The application and use of Compact Thermal Models of single chip electronic packages in a board level environment has seen limited study. In particular the influence of board thermal characteristics on the predictive accuracy of CTMs generated independently of the board is not well understood. A systematic study of the influence of board conduction on the predictive accuracy of Compact Thermal Models of BGA and CPGA package styles was performed. For the CPGA special attention was given to the socket model. The compact model parameters were extracted using a standard optimization procedure to best reproduce the junction temperature and heat flows computed for the detailed models exposed to a reduced set of heat transfer coefficients on the prime lumped areas. Various resistance network topologies were assessed for each package style. Detailed (FE) isotropic board models with conductivities spanning three orders of magnitude were created to test the influence of board conductivity on CTM accuracy. The board models included fully detailed isotropic models of the 1S0P and 1S2P JEDEC standard thermal test boards. Compact models were deployed on these detailed board models with the board surfaces exposed to a wide variety of convective cooling conditions to test the limits of the package models. Results were compared to equivalent fully-detailed (FE) package models on detailed board models. The predictive capability of the optimized topologies was strongly correlated to the existence of strong local temperature gradients in the board, whenever the sensitivity to board heat conduction was high. When the board is poorly conducting ($k=1$ W/m-K), board heat conduction is too low to matter. When the board is highly conducting ($k=100$ W/m-K), the high conductivity smooths out local board temperature gradients, and the CTMs are accurate even though the board heat flow path is dominant. For intermediate board thermal conductivities of order 10 W/m-K, the heat conduction into the board is significant enough to cause sensitivity, and the conductivity is low enough to allow significant board temperature gradients. These conditions significantly degrade the predictive accuracy of CTMs. In general, optimized network topologies that included shunt resistances were found to predict the junction temperature to within 5% to 8% over a wide range of board conductivities, while star-shaped networks were generally only accurate to 10-15%.

Key Words: compact thermal models, CPGA, BGA, fully detailed model, printed circuit board, network topology

Nomenclature

AR	area ratio
h	average heat transfer coefficient [W/m ² K]
R	thermal link [°C /W]
T	temperature [°C]

Subscripts

bi	bottom in
bo	bottom out
b1, b2	split bottom areas
edge(s)	edge of board
CTM / FDM	compact thermal model of package, fully detailed model of board
FDM / FDM	fully detailed model of package and board
J	junction
s	side
t	top
t1, t2	split top areas
ti	top in
to	top out
t/b, board	top/bottom of board

Introduction

The justification for and procedures used in deriving Compact Thermal Models (CTM) of electronic packages are well documented in the literature. A successful CTM is a few-parameter representation of a complex thermal structure that is able to simulate the thermal response of a package (junction temperature, dominant heat flows) with adequate fidelity in a larger system model. The “system” is generally understood to mean the package, its printed circuit board and surrounding heat dissipating components, and the surrounding boards and peripheral components enclosed in some type of chassis or enclosure. The concept of “embedded simplified or compact models” that synthesize the response of a complex sub-component of a larger system in a computationally efficient manner is commonly used in fields as diverse as electrical simulation of circuits and simulation of failure in complex mechanical structures. In the thermal simulation of electronic systems, in particular single chip packages, the most studied and perhaps most promising methodology is the statistical parameter extraction method popularized in Europe by project DELPHI [1]. In this method, a Finite Element (FE) or Finite Difference (FD) detailed model of the package is used to

generate thermal response data (junction temperature, primary heat flows) for a statistically meaningful set of thermal boundary conditions applied to a relatively few distinct external surfaces. Most commonly, the thermal boundary conditions applied to the surfaces are Robin type conditions, where the ratio of the total heat flux through the surface to the average temperature of the surface above some reference, the so-called heat transfer coefficient, is specified. By an optimization procedure, the parameters of the compact model are extracted such as to globally minimize the error in predicting the response data. In the parameter extraction process, the same heat transfer coefficient boundary conditions applied to the detailed model surfaces are systematically applied to the compact model, and the thermal response predicted by the compact model is compared to the detailed model to determine the predictive accuracy. Significantly fewer studies have been conducted to investigate the predictive accuracy of CTM's when they are embedded in models of air-cooled, conductive, populated printed circuit boards. Many theoretical and practical issues are yet to be addressed. This paper will report on one important issue, which is the influence of board thermal characteristics on the predictive accuracy of CTMs generated using the procedure described above.

Previous Work

The methodology for generating compact models using this procedure has been exhaustively reported, see for example [1-6]. The most significant result found thus far is that it is possible to create optimized few parameter CTMs that are mostly independent of the magnitude and distribution of the heat transfer coefficient boundary conditions applied to the external surfaces. External heat transfer coefficients are commonly used to simulate the convective heat transfer paths, but in addition, Aranyosi et al [6] found that external equivalent heat transfer coefficients can be successfully used to simulate the heat transfer path through surface mount package leads to their board, thus allowing the generation of optimized CTMs independently of their board. When embedded in a system model, the external heat transfer coefficient links on the package surfaces are removed, and the surface nodes are linked to the system detailed model. On the air side, the system model is usually one of two types: a non-conjugate model in which the convective cooling flow effects are represented by heat transfer coefficients linked to the package CTM and board surfaces [6], or a conjugate CFD model, in which the energy equation of the flow is discretized and the discrete air-side nodes are attached directly to the solid surface nodes [2,3,4]. On the board side, CTM surface nodes are linked directly to the mating surface nodes of the board discretized model irrespective of the approach used in modeling the flow. Simple one resistance junction- to-ambient models were embedded in a CFD simulation of the flow in [2]. The accuracy of this approach was not discussed in comparison to experiment or fully detailed models. A star-shaped thermal resistance network and a uniform property block representation were tested in a CFD simulation of a natural convection flow in [3]. CTM's in a system level analysis were compared to a FDM of the entire system and

experiments. Reported accuracy was 12-13% from FDM to experiment and 7.5-13% for CTM to FDM comparisons. A more complex compact thermal model was embedded in a CFD model of a forced convection cooling scenario with a heat sink in [4]. The resistance networks were approximated within the system level model by a mixture of high conductivity blocks with surface resistances chosen at the interfaces. The reported accuracy was within 5% for the optimized network and 2-20% for the two resistance characterization.

When board conduction is important, the board thermal characteristics may influence the predictive accuracy of the CTM. As shown in [5], the temperature distribution of the mating package/board surfaces is highly dependent on the heat flow distribution in the board, which is in turn determined by the method in which heat is removed from the board surfaces. Because the CTM represents the package exterior by a relatively few surfaces, the surface temperature node thus represents a spatially averaged temperature. Since the package thermal response data was generated by applying an equivalent and uniform heat transfer coefficient to the package external surfaces as in [5], it is presumed that the process of encapsulating surface temperature variations in the surface averaged node temperature data used to derive the CTM is robust. It must be at least robust enough to ensure predictive accuracy when the CTM is coupled to the system level detailed board model that may produce significantly different surface temperature variations. This issue is the subject of the present paper. The work reported here is a continuation of the study initiated in [6] and the reader is referred to this earlier paper for further background details.

Package Compact Model Generation

A ceramic pin grid array (CPGA) and a ball grid array (BGA) package were selected as representative package families for investigating board level issues. Cross-sections of both packages are shown in Figs. 1 and 2. The CPGA was a derivative of that reported in [6] with some modifications. The original model used an air/pin smear in order for comparison with experimental procedures. On a board level, this air is not present and the pins are instead immersed in a plastic socket. A copper lid was also added to the model as specified by [7]. These changes necessitated the re-evaluation of the CTM topology. The detailed model was not experimentally validated; it is assumed to be comparably precise to the original, validated case.

The ball grid array package specifications adopted duplicated the package model utilized in a recent JEDEC standardization exercise [8,9]. The detailed BGA model faithfully reproduced the geometry that was experimentally validated in [9]. This provided an experimentally validated test package and board and a detailed report of all thermo-physical properties and assumptions. While the detailed model was only indirectly validated, it was precise enough for the purpose of assessing the board conduction issues, which was the principle motivation of the study. In both package models, contact resistances were placed so that each model can be properly "tuned" with experimental data once these validations are performed

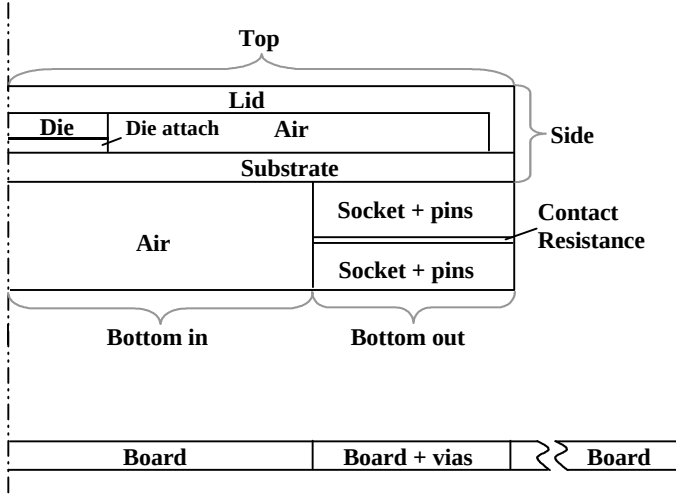


Figure 1: Cross-section view of CPGA package

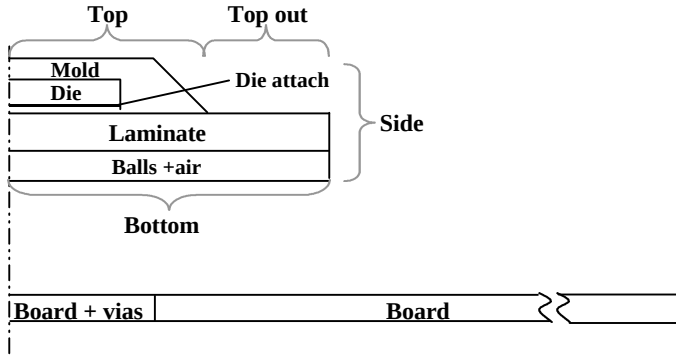


Figure 2: Cross-section view of BGA package

For this study, the design of experiments approach reported in [5] and [6] was used. Eight non-redundant sets of heat transfer coefficient boundary conditions were obtained and applied to the fully detailed model to generate the thermal response data needed to derive the compact thermal model. The shunted CTMs were optimized to predict junction temperature to within 5% and heat flows to within 10% of that found with a fully detailed model while minimizing the following cost function [10]:

$$F = \frac{1}{BC} \sum_{k=1}^{BC} \left[V \left(\frac{T_c(k) - T_f(k)}{T_f(k) - T_{amb}} \right)^2 + \frac{W}{N} \sum_{i=1}^N \left(\frac{\Phi_{i,f}(k) - \Phi_{i,c}(k)}{\Phi_{total}} \right)^2 \right]$$

The star-shaped CTM's were only constrained to 20% errors on the junction temperature and 25% errors on the heat fluxes. For all cases, the primary lumped areas were sometimes split with the area ratio as an optimized parameter. The optimization procedure was carried out in a custom spreadsheet based code described in [6].

The topologies for the shunted networks are iteratively found within the optimization routine. Every possible resistance is started with a finite value. Once the optimum values are found, links with very high resistances are closed

and the optimization procedure is again initiated. This iteration loop is run until a resistance network is found that meets the minimum error requirements, minimizes the cost function for that iteration, and contains a reasonable number of links (typically eight or less). The final (shunted) topology is by no means an arbitrary or predetermined shape.

Several compact thermal models were derived for each package. For the CPGA, two scenarios were considered. First, the package model was cut above the contact resistance layer and modeled to only include the portion of the socket that comes in direct contact with the pins of the chip package. An optimized network topology including shunt resistances was obtained using the iterative process previously described. The derivation included the division of the top area into two parts with the area ratio as an optimized parameter. A simple star-shaped network was also derived again allowing for the top area to be split and optimized. A second case was modeled including the entire socket. The shunted network for this case was forced to have the same topology as the partial socket case but the area ratio and resistance values were allowed to vary. This shunted topology for both cases is shown in Figure 3.

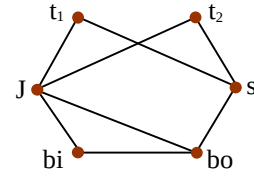


Figure 3: CTM network topology for the CPGA chip package with full and partial socket.

The BGA package was analyzed according to three scenarios. Optimized shunted and star-shaped networks were developed for each scenario. Case One was a four node network without any area splitting. Case Two allowed the top area to be split and optimized in the same manner as the CPGA. Case Three split the bottom area into two parts but the area ratio was fixed so that the represented areas were equal to those portions of the board with and without thermal vias. This case was of particular interest because it allowed two paths of heat into the board whereas the previous cases only allowed one. The two latter cases also allowed for comparison between fixed area splitting and the optimization of the area ratio. The three shunted topologies for the BGA chip package are shown in Figure 4.

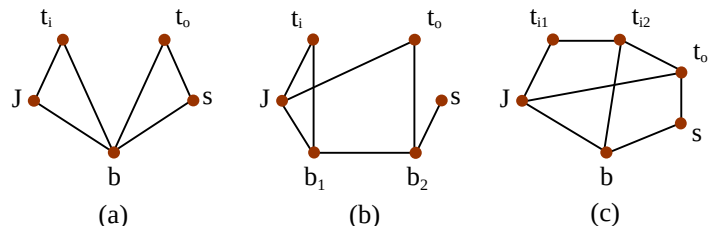


Figure 4: CTM topologies for (a) Four Node BGA, (b) Split Bottom BGA, and (c) Split Top BGA.

Board Level Modeling

Fully detailed models of the board were developed. Smeared (or averaged) anisotropic conductivities were calculated using JEDEC dimensions and specifications for the standard 1S0P and 1S2P thermal test boards for each package style [11,12]. In-plane conductivities were approximated with parallel weighting while series weighting was used for the through-plane values. Since the heat predominantly flows in the direction of the traces, the accuracy of the in-plane conductivity in the direction perpendicular to the traces was not crucial. For this reason, the in-plane conductivity was assumed constant and was set to the value of the weighted average found in the direction parallel to the traces. The values of these conductivities are shown below in Table 1.

	1S0P	1S2P
In-plane (W/m-K)	17.40	26.19
Through-plane (W/m-K)	0.366	0.396

Table 1: Averaged board conductivities for JEDEC standard thermal test boards

Boards with isotropic properties were developed as well. The range of values was chosen to cover three orders of magnitude ($k = 2, 10$, and 100 W/m-K) in order to evaluate dependence of accuracy on board conductivity.

A fully detailed package model was placed on each detailed board model. A set of extreme heat transfer coefficient boundary condition sets were applied to the board, to produce extreme variations in the board heat flow paths, as previously suggested in [5]. A typical detailed model is shown in Fig. 5. These cases (FDM / FDM) are assumed to be the “exact” cases. The fully detailed models were replaced with the derived CTM’s for each package. The finite difference code used in the study had the advantageous feature of allowing arbitrary surface links to be attached to surface nodes, including the possibility of shorting the surface links to each other. This latter feature was invaluable in placing shunted CTMs on the board models. An example of a CTM model deployed on a board model is shown in Fig. 6. These cases (CTM / FDM) were tested for the same boundary conditions and compared with the “exact” results.

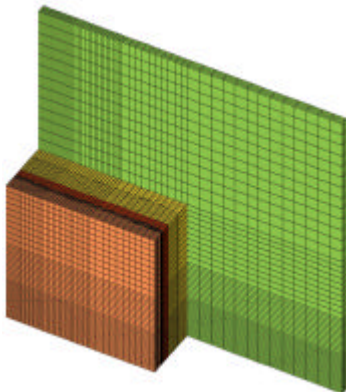


Figure 5: Fully detailed model of board and CPGA (FDM / FDM)

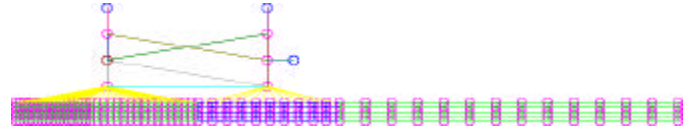


Figure 6: Compact thermal model of CPGA on fully detailed model of the board (CTM / FDM)

The extreme boundary condition sets used to compare the two models are shown in Table 2. Convection is placed on the sides (h_s) and top (h_t) of the package as in the package level analysis. The bottom of the package is attached to the board so the convection link is replaced by a direct link to the board nodes. Convection is placed on the top and bottom of the board ($h_{t/b, \text{board}}$) and on the board edges ($h_{\text{edge}(s)}$).

BC Set	h_t	h_s	$h_{t/b, \text{board}}$	$h_{\text{edge}(s)}$
A	1000	1000	5	0
B	800	250	50	0
C	5	5	1000	0
D	5	5	0	1000
E	5	5	0	1000

Table 2: Boundary condition sets for the board level models.

Each set was chosen to test a different extreme of the board level model. Case A tested the response when the primary heat path was through the package. Case B was chosen as an arbitrary scenario. Case C was established to simulate cooling through the top and bottom areas of the board. Test case D was a non-symmetric edge cooling case where convection was only applied to one edge of the board. Case E primarily removed heat through all four board edges.

Results and Discussion

The shunted CTM / FDM models were generally found to predict the junction temperature to within 3-5% of the FDM / FDM case for boundary condition sets A, B, and C. Cases A and B minimize the board conduction and hence do not test the board conduction sensitivity. Case C promotes conduction into the board in a radially symmetric pattern from the package, because the cooling on the top and bottom board surfaces is uniform and equal. For the edge cooling scenarios D and E however, the errors were as high as 11.6%, and were generally higher for asymmetric edge cooling, case D, than for symmetric edge cooling, case E. The errors appear to be strongly correlated to the magnitude of the temperature gradients in the board in the vicinity of the package. By representing the bottom of the package as one or two nodes, the CTM / FDM model cannot accurately distribute the heat flow approaching the one or two nodes to match the actual surface averaged temperatures that the package faces in those cases. The only scenario that consistently returned accurate results for boundary conditions D and E were the CPGA cases in which only a portion of the socket was included in the CTM. The portion of the socket that remains connected to

the fully detailed model smoothes the temperature contours considerably so that the bottom of the partial socket model is exposed to relatively small variation in temperature. The average error for the shunted networks were 3.4% for the BGA and 2.0% for the CPGA

Not surprisingly, the star-shaped CTM / FDM cases had much higher errors for all of the cases. However, the partial socket CPGA star-shaped compact thermal model was only in error by an average of 2.3%. The full socket CPGA star model was only accurate to an average of 10.3%. The accuracy of star-shaped models for the BGA package was uniformly unacceptable for all cases. The average error was 20.7% for with a maximum error of 66.1%.

It is important to note that the temperature contours on the board were very similar for the successful FDM / FDM and the CTM / FDM cases. This can be seen in Figure 7 for one of the runs. The package geometry is not represented in the CTM / FDM case as it has been removed and replaced with the CTM of the package. This reinforces the hypothesis that in order for a CTM to be successful, it must be able to distribute the heat flow to the various branches not only to satisfy an overall energy balance but also to return reasonably accurate average surface temperatures over the lumped areas.

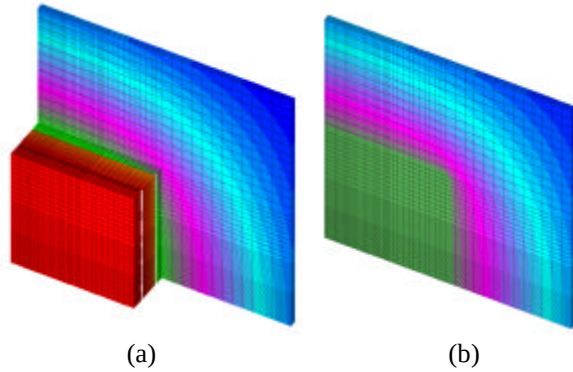


Figure 7: Temperature contours for CPGA with a full socket, shunted network, and isotropic board conductivity ($k = 10$ W/m-K) subjected to Boundary Condition Set E. Both cases (a) FDM / FDM and (b) CTM / FDM (package geometry not represented) are shown.

The relative errors for the CPGA package are shown in Figs. 8 and 9 and for the BGA package in Figs. 10 and 11. In Figs. 8 and 9, all five boundary condition sets are shown for the four scenarios. Results for every board conductivity are plotted. In Figs. 10 and 11, all five boundary condition sets are present for each of the six scenarios.

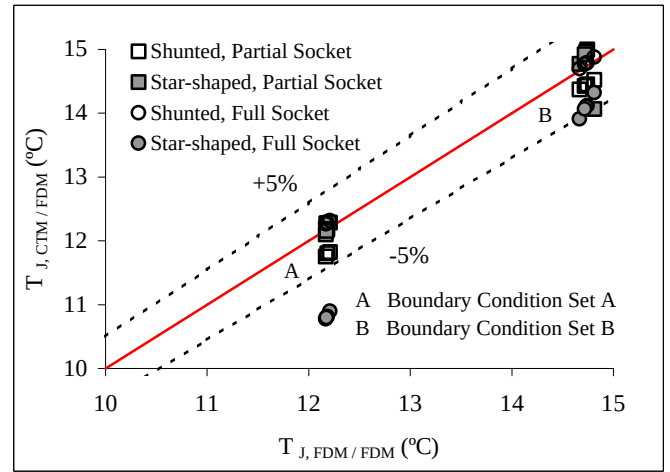


Figure 8: Compact thermal model vs. detailed model for CPGA, Boundary conditions A and B.

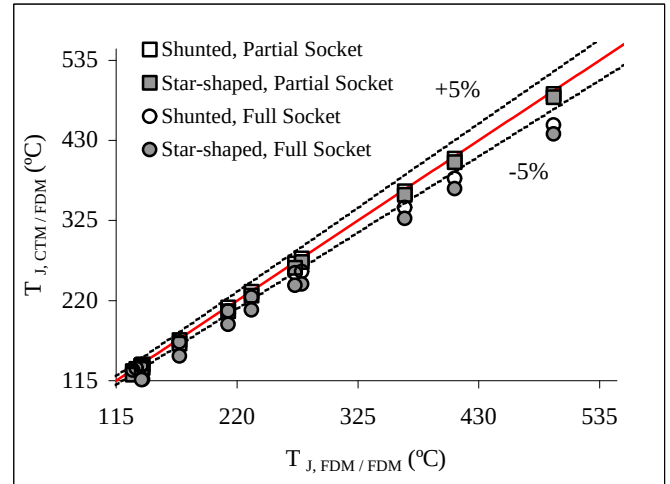


Figure 9: Compact thermal model vs. detailed model for CPGA, Boundary conditions C, D, and E.

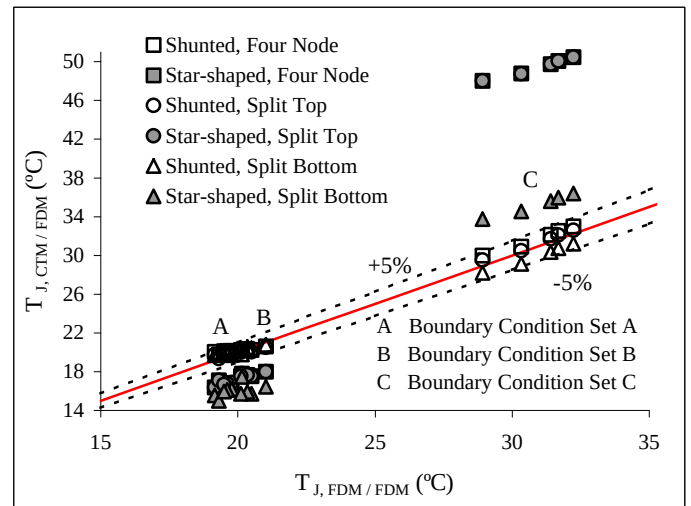


Figure 10: Compact thermal model vs. detailed model for BGA, Boundary conditions A, B and C.

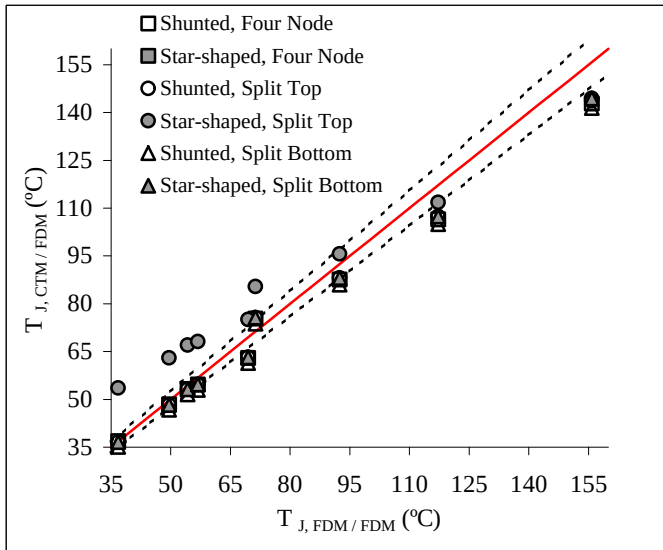


Figure 11: Compact thermal model vs. detailed model for BGA, Boundary conditions D and E.

As previously mentioned, the inclusion of the entire socket in the CPGA compact thermal models actually worsened the predictive capability of the networks in most cases, Fig. 9. This was especially apparent in the edge cooling scenarios (boundary condition sets D and E) for both the star-shaped and shunted networks.

Changes in the topology for the shunted networks (BGA) did not produce any dramatic changes in accuracy, Fig. 11. While the split top case produced the least average error (2.8%), the split bottom (4.1%) and basic four node (3.5%) produced equally reasonable accuracy. The differences in the split bottom and split top cases do suggest that optimizing the area ratio is a better approach to properly characterize the package than setting a fixed area ratio. Considering that the split bottom case returned the highest average error, there seems no reason to believe that adding a second heat path into the board is necessarily beneficial if it does not account for a significant percentage of heat flow into the board.

Careful interpretation of Figs. 12 and 13 shows an absolutely distinguishable correlation between the predictive accuracy of the CTM and the magnitude of the board temperature gradients to which it is exposed. At first glance, there does not appear to be any relation between accuracy and board conductivity. However, both package styles demonstrate the greatest error at an intermediate thermal conductivity of 10 W/m-K. Our interpretation is as follows. At a low thermal conductivity of 2, the board conduction path is becoming important but not dominant. The board thermal gradients are high, but the overall heat conduction to the board is low, hence, overall predictive accuracy is not affected by large errors in the distribution of the heat flow into the board from the package. For a board thermal conductivity of 100 W/m-K, the overall heat conduction into the board is high, but the spatial temperature gradients are low, due to the smoothing effect of the highly conductive board. This is especially obvious for the BGA package, Fig. 13. At an intermediate value of 10 W/m-K, conduction into the board is significant enough to affect the overall predictive accuracy. However,

the conductivity is still sufficiently low to produce large local temperature gradients in the board for cases D and E. Thus, it is clear that local board thermal characteristics have a detrimental affect on the CTM predictive capability, whenever the magnitude of the conduction into the board is significant enough to make the model sensitive to the board heat flow paths, and the local board temperature gradients are high. Effective board thermal conductivities on the order of 10 W/m-K appear to induce such effects for both the CPGA and BGA packages. It is useful to note in closing that the effective thermal conductivities of the 1S0P and 1S2P JEDEC standard boards are roughly of this order of magnitude (Table 1.) and inspection of Figs. 12 and 13 shows that the CTM predictions are indeed sensitive to accurate modeling of the board conduction, as can be seen for cases D and E.

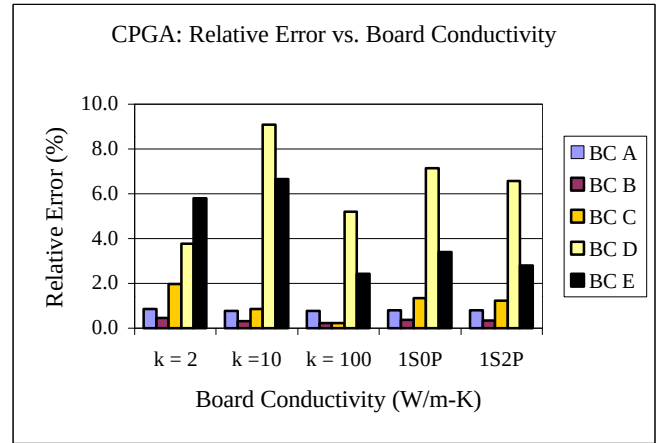


Figure 12: Relative error for CPGA package (full socket, shunted network) for varied board conductivities and boundary condition set.



Figure 13: Relative error for BGA package (split top, shunted network) for varied board conductivities and boundary condition set.

Conclusions

1. When heat conduction into the board is significant, the sensitivity of the CTM predictive accuracy to proper modeling of the heat flow paths into the board increases.

2. In all cases, whether or not board conduction is important, shunted topologies outperform star-shaped topologies. Shunted networks calculated junction temperatures within 5% to 8%, depending on board conductivity. Star-shaped networks generally predicted junction temperature to within 15-20% but had maximum error of over 65%.
3. For optimized shunted models, the predictive accuracy degraded when the model was exposed to large local board temperature gradients, and when the magnitude of the board conduction was significant. These conditions exist when the board effective isotropic conductivity is on the order of 10 W/m-K.
4. There was no distinguishable variation in the accuracy of the different shunted network topologies, although splitting both the upper and lower package areas was beneficial, especially when the area ratio was an optimizable parameter.

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